

Y

San Diego, CA | (858) 282-8657 | wuyou.nsfz@outlook.com | github.com/yow008

SUMMARY

Motivated master's student majoring in Computer Engineering, experienced in developing cross-platform web applications, with strong programming skills and effective communication ability. Actively looking for a summer internship in 2025

EDUCATION

University of California, San Diego

Master of Science in Electrical Engineering (Computer Engineering)

Expected December 2025

Bachelor of Science in Computer Engineering | GPA: 3.706

December 2023

Relevant Courses: Advanced Data Structures, Design and Analysis of Algorithms, Software Engineering, Data Networks, Database System Principles

TECHNICAL SKILLS

- **Programming Languages:** JavaScript, CSS, HTML, Java, C, C++, Python, SQL
- **Software Tools and Web Frameworks:** Unix/Linux, Git, Visual Studio Code, Jira, JSON, React, Spring

WORK EXPERIENCE

NRIET

June 2024 – Present

Software Engineering Intern

Nanjing, China

- Developed a software in C to manage state changes of microcontrollers in an interlocking system
- Implemented multithreading to handle data input from GPIO ports and output via CAN bus, reducing processing time
- Debugged and resolved timing conflicts in communications across 20 boards by utilizing timers for precise time allocation, ensuring the reliability with 0 package loss

Dr. Rose Yu's Spatiotemporal Machine Learning Lab, UC San Diego

June – October 2022

Application Developer

San Diego, CA

- Developed a web app (limo-aimd.com) for pharmaceutical professionals to interact with and evaluate a molecule generation model, enabling the lab to present the model at conferences
- Built the app using React, collaborating closely with the lab to align the design match their goals and integrate the app seamlessly with the SQLite database and APIs
- Visualized molecular 3D structures using NGL and enhanced the results page by leveraging MUI components, resulting in over 370 trials of the model

PROGRAMMING PROJECTS

Cache Simulator

May – June 2024

- Simulated a two-level cache structure in C++, including an instruction cache and a data cache on level 1, with an option to set level 2 cache as inclusive or non-inclusive
- Implemented a least recently used (LRU) replacement policy and a next line prefetcher for the two-level cache to enhance cache performance and efficiency

Data Science Project on Housing Sales Price and Household Income

February – March 2023

- Utilized Python in a data science project to analyze the correlation between average housing sales price and median household income in San Diego County
- Performed data cleaning and visualization using Python libraries including pandas, seaborn, NumPy, and Matplotlib

CookEZ Recipe App

October – December 2021

- Developed a recipe application using HTML, JavaScript, and Spoonacular API following Agile principles
- Designed and implemented the user interface using Figma and CSS, simplifying the presentation of the ingredient list and cooking mode pages

LEADERSHIP & EXTRACURRICULAR ACTIVITY

Business & Financial Services, UC San Diego

December 2023 – March 2024

Bike Courier

San Diego, CA

- Delivered packages to labs and offices across campus riding an electronic tricycle in various weather conditions

UC San Diego Computer Science and Engineering Department (CSE)

January – March 2023

Tutor

San Diego, CA

- Led discussion sessions for groups of 20 students in the absence of a teaching assistant, providing detailed explanations and addressing questions to enhance understanding
- Guided students in learning Android Studio and Java during lab sessions, and evaluated their code and solutions